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compact3

java.lang.management

Class ManagementFactory

java.lang.Object
    java.lang.management.ManagementFactory

```
public class ManagementFactory
extends Object
```

The `ManagementFactory` class is a factory class for getting managed beans for the Java platform. This class consists of static methods each of which returns one or more *platform MXBeans* representing the management interface of a component of the Java virtual machine.

Platform MXBeans

A platform MXBean is a *managed bean* that conforms to the JMX Instrumentation Specification and only uses a set of basic data types. A JMX management application and the platform MBeanServer can interoperate without requiring classes for MXBean specific data types. The data types being transmitted between the JMX connector server and the connector client are open types and this allows interoperation across versions. See the specification of MXBeans for details.

Each platform MXBean is a `PlatformManagedObject` and it has a unique `ObjectName` for registration in the platform `MBeanServer` as returned by by the `getObjectName` method.

An application can access a platform MXBean in the following ways:

1. Direct access to an MXBean interface

- Get an MXBean instance by calling the `getPlatformMXBean` or `getPlatformMXBeans` method and access the MXBean locally in the running virtual machine.

- Construct an MXBean proxy instance that forwards the method calls to a given `MBeanServer` by calling the `getPlatformMXBean(MBeanServerConnection, Class)` or `getPlatformMXBeans(MBeanServerConnection, Class)` method. The `newPlatformMXBeanProxy` method can also be used to construct an MXBean proxy instance of a given `ObjectName`. A proxy is typically constructed to remotely access an MXBean of another running virtual machine.

2. Indirect access to an MXBean interface via MBeanServer

- Go through the platform `MBeanServer` to access MXBeans locally or a specific `MBeanServerConnection` to access MXBeans remotely. The attributes and operations of an MXBean use only *JMX open types* which include basic data types, `CompositeData`, and `TabularData` defined in `OpenType`. The mapping is specified in the MXBean specification for details.

The `getPlatformManagementInterfaces` method returns all management interfaces supported in the Java virtual machine including the standard management interfaces listed in the tables below as well as the management interfaces extended by the JDK implementation.

A Java virtual machine has a single instance of the following management interfaces:

Management Interface	ObjectName
`ClassLoadingMXBean`	`java.lang:type=ClassLoading`
`MemoryMXBean`	`java.lang:type=Memory`
`ThreadMXBean`	`java.lang:type=Threading`
`RuntimeMXBean`	`java.lang:type=Runtime`
`OperatingSystemMXBean`	`java.lang:type=OperatingSystem`
`PlatformLoggingMXBean`	`java.util.logging:type=Logging`

A Java virtual machine has zero or a single instance of the following management interfaces.

Management Interface	ObjectName
`CompilationMXBean`	`java.lang:type=Compilation`

A Java virtual machine may have one or more instances of the following management interfaces.

Management Interface	ObjectName
`GarbageCollectorMXBean`	`java.lang:type=GarbageCollector,name=`*collector's name*
`MemoryManagerMXBean`	`java.lang:type=MemoryManager,name=`*manager's name*

MemoryPoolMXBean	java.lang:type=MemoryPool,name=*pool's name*
BufferPoolMXBean	java.nio:type=BufferPool,name=*pool name*

Since:

1.5

See Also:

JMX Specification, Ways to Access Management Metrics, MXBean

Field Summary

Fields

Modifier and Type	Field	Description
static String	CLASS_LOADING_MXBEAN_NAME	String representation of the ObjectName for the ClassLoadingMXBean.
static String	COMPILATION_MXBEAN_NAME	String representation of the ObjectName for the CompilationMXBean.
static String	GARBAGE_COLLECTOR_MXBEAN_DOMAIN_TYPE	The domain name and the type key property in the ObjectName for a GarbageCollectorMXBean.
static String	MEMORY_MANAGER_MXBEAN_DOMAIN_TYPE	The domain name and the type key property in the ObjectName for a MemoryManagerMXBean.
static String	MEMORY_MXBEAN_NAME	String representation of the ObjectName for the MemoryMXBean.
static String	MEMORY_POOL_MXBEAN_DOMAIN_TYPE	The domain name and the type key property in the ObjectName for a MemoryPoolMXBean.
static String	OPERATING_SYSTEM_MXBEAN_NAME	String representation of the ObjectName for the OperatingSystemMXBean.
static String	RUNTIME_MXBEAN_NAME	String representation of the ObjectName for the RuntimeMXBean.

static String	THREAD_MXBEAN_NAME	String representation of the ObjectName for the ThreadMXBean.

Method Summary

Concrete Methods

Modifier and Type	Method
static ClassLoadingMXBean	getClassLoadingMXBean()
static CompilationMXBean	getCompilationMXBean()
static List<GarbageCollectorMXBean>	getGarbageCollectorMXBeans()
static List<MemoryManagerMXBean>	getMemoryManagerMXBeans()
static MemoryMXBean	getMemoryMXBean()
static List<MemoryPoolMXBean>	getMemoryPoolMXBeans()
static OperatingSystemMXBean	getOperatingSystemMXBean()
static Set<Class<? extends PlatformManagedObject>>	getPlatformManagementInterfac

static **MBeanServer**	**getPlatformMBeanServer**()
static <T extends **PlatformManagedObject**> T	**getPlatformMXBean**(**Class**<T> mx
static <T extends **PlatformManagedObject**> T	**getPlatformMXBean**(**MBeanServer** **Class**<T> mxbeanInterface)
static <T extends **PlatformManagedObject**> **List**<T>	**getPlatformMXBeans**(**Class**<T> m
static <T extends **PlatformManagedObject**> **List**<T>	**getPlatformMXBeans**(**MBeanServe** **Class**<T> mxbeanInterface)
static **RuntimeMXBean**	**getRuntimeMXBean**()
static **ThreadMXBean**	**getThreadMXBean**()

static <T> T	newPlatformMXBeanProxy(MBeanS String mxbeanName, Class<T> m

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Detail

CLASS_LOADING_MXBEAN_NAME

public static final String CLASS_LOADING_MXBEAN_NAME

String representation of the ObjectName for the ClassLoadingMXBean.

See Also:

Constant Field Values

COMPILATION_MXBEAN_NAME

public static final String COMPILATION_MXBEAN_NAME

String representation of the ObjectName for the CompilationMXBean.

See Also:

Constant Field Values

MEMORY_MXBEAN_NAME

public static final String MEMORY_MXBEAN_NAME

String representation of the ObjectName for the MemoryMXBean.

See Also:

Constant Field Values

OPERATING_SYSTEM_MXBEAN_NAME

```
public static final String OPERATING_SYSTEM_MXBEAN_NAME
```

String representation of the `ObjectName` for the `OperatingSystemMXBean`.

See Also:

Constant Field Values

RUNTIME_MXBEAN_NAME

```
public static final String RUNTIME_MXBEAN_NAME
```

String representation of the `ObjectName` for the `RuntimeMXBean`.

See Also:

Constant Field Values

THREAD_MXBEAN_NAME

```
public static final String THREAD_MXBEAN_NAME
```

String representation of the `ObjectName` for the `ThreadMXBean`.

See Also:

Constant Field Values

GARBAGE_COLLECTOR_MXBEAN_DOMAIN_TYPE

```
public static final String GARBAGE_COLLECTOR_MXBEAN_DOMAIN_TYPE
```

The domain name and the type key property in the `ObjectName` for a `GarbageCollectorMXBean`. The unique `ObjectName` for a `GarbageCollectorMXBean` can be formed by appending this string with "`,name=`*collector's name*".

See Also:

Constant Field Values

MEMORY_MANAGER_MXBEAN_DOMAIN_TYPE

```
public static final String MEMORY_MANAGER_MXBEAN_DOMAIN_TYPE
```

The domain name and the type key property in the ObjectName for a MemoryManagerMXBean. The unique ObjectName for a MemoryManagerMXBean can be formed by appending this string with ",name=*manager's name*".

See Also:

Constant Field Values

MEMORY_POOL_MXBEAN_DOMAIN_TYPE

```
public static final String MEMORY_POOL_MXBEAN_DOMAIN_TYPE
```

The domain name and the type key property in the ObjectName for a MemoryPoolMXBean. The unique ObjectName for a MemoryPoolMXBean can be formed by appending this string with ,name=*pool's name*.

See Also:

Constant Field Values

Method Detail

getClassLoadingMXBean

```
public static ClassLoadingMXBean getClassLoadingMXBean()
```

Returns the managed bean for the class loading system of the Java virtual machine.

Returns:

a ClassLoadingMXBean object for the Java virtual machine.

getMemoryMXBean

```
public static MemoryMXBean getMemoryMXBean()
```

Returns the managed bean for the memory system of the Java virtual machine.

Returns:

a MemoryMXBean object for the Java virtual machine.

getThreadMXBean

```
public static ThreadMXBean getThreadMXBean()
```

Returns the managed bean for the thread system of the Java virtual machine.

Returns:

a `ThreadMXBean` object for the Java virtual machine.

getRuntimeMXBean

```
public static RuntimeMXBean getRuntimeMXBean()
```

Returns the managed bean for the runtime system of the Java virtual machine.

Returns:

a `RuntimeMXBean` object for the Java virtual machine.

getCompilationMXBean

```
public static CompilationMXBean getCompilationMXBean()
```

Returns the managed bean for the compilation system of the Java virtual machine. This method returns `null` if the Java virtual machine has no compilation system.

Returns:

a `CompilationMXBean` object for the Java virtual machine or `null` if the Java virtual machine has no compilation system.

getOperatingSystemMXBean

```
public static OperatingSystemMXBean getOperatingSystemMXBean()
```

Returns the managed bean for the operating system on which the Java virtual machine is running.

Returns:

an `OperatingSystemMXBean` object for the Java virtual machine.

getMemoryPoolMXBeans

```
public static List<MemoryPoolMXBean> getMemoryPoolMXBeans()
```

Returns a list of `MemoryPoolMXBean` objects in the Java virtual machine. The Java virtual machine can have one or more memory pools. It may add or remove memory pools during execution.

Returns:

a list of MemoryPoolMXBean objects.

getMemoryManagerMXBeans

```
public static List<MemoryManagerMXBean> getMemoryManagerMXBeans()
```

Returns a list of MemoryManagerMXBean objects in the Java virtual machine. The Java virtual machine can have one or more memory managers. It may add or remove memory managers during execution.

Returns:

a list of MemoryManagerMXBean objects.

getGarbageCollectorMXBeans

```
public static List<GarbageCollectorMXBean> getGarbageCollectorMXBeans()
```

Returns a list of GarbageCollectorMXBean objects in the Java virtual machine. The Java virtual machine may have one or more GarbageCollectorMXBean objects. It may add or remove GarbageCollectorMXBean during execution.

Returns:

a list of GarbageCollectorMXBean objects.

getPlatformMBeanServer

```
public static MBeanServer getPlatformMBeanServer()
```

Returns the platform MBeanServer. On the first call to this method, it first creates the platform MBeanServer by calling the MBeanServerFactory.createMBeanServer method and registers each platform MXBean in this platform MBeanServer with its ObjectName. This method, in subsequent calls, will simply return the initially created platform MBeanServer.

MXBeans that get created and destroyed dynamically, for example, memory pools and managers, will automatically be registered and deregistered into the platform MBeanServer.

If the system property javax.management.builder.initial is set, the platform MBeanServer creation will be done by the specified MBeanServerBuilder.

It is recommended that this platform MBeanServer also be used to register other application managed beans besides the platform MXBeans. This will allow all MBeans to be published through the same MBeanServer and hence

allow for easier network publishing and discovery. Name conflicts with the platform MXBeans should be avoided.

Returns:

`the platform MBeanServer; the platform MXBeans are registered into the platform MBeanServer at the first time this method is called.`

Throws:

`SecurityException - if there is a security manager and the caller does not have the permission required by MBeanServerFactory.createMBeanServer().`

See Also:

`MBeanServerFactory, MBeanServerFactory.createMBeanServer()`

newPlatformMXBeanProxy

```
public static <T> T newPlatformMXBeanProxy(MBeanServerConnection connection,
                                           String mxbeanName,
                                           Class<T> mxbeanInterface)
                                    throws IOException
```

Returns a proxy for a platform MXBean interface of a given MXBean name that forwards its method calls through the given `MBeanServerConnection`.

This method is equivalent to:

```
Proxy.newProxyInstance(mxbeanInterface.getClassLoader(),
new Class[] { mxbeanInterface }, handler)
```

where `handler` is an `InvocationHandler` to which method invocations to the MXBean interface are dispatched. This `handler` converts an input parameter from an MXBean data type to its mapped open type before forwarding to the `MBeanServer` and converts a return value from an MXBean method call through the `MBeanServer` from an open type to the corresponding return type declared in the MXBean interface.

If the MXBean is a notification emitter (i.e., it implements `NotificationEmitter`), both the `mxbeanInterface` and `NotificationEmitter` will be implemented by this proxy.

Notes:

1. Using an MXBean proxy is a convenience remote access to a platform MXBean of a running virtual machine. All method calls to the MXBean proxy are forwarded to an `MBeanServerConnection` where `IOException` may be thrown when the communication problem occurs with the connector server. An application remotely accesses the platform MXBeans using proxy should prepare to catch `IOException` as if accessing with the `MBeanServerConnector` interface.

2. When a client application is designed to remotely access MXBeans for a running virtual machine whose version is different than the version on which the application is running, it should prepare to catch `InvalidObjectException` which is thrown when an MXBean proxy receives a name of an enum constant which is missing in the enum class loaded in the client application.
3. `MBeanServerInvocationHandler` or its `newProxyInstance` method cannot be used to create a proxy for a platform MXBean. The proxy object created by `MBeanServerInvocationHandler` does not handle the properties of the platform MXBeans described in the class specification.

Type Parameters:

`T - an mxbeanInterface type parameter`

Parameters:

`connection - the MBeanServerConnection to forward to.`

`mxbeanName - the name of a platform MXBean within connection to forward to. mxbeanName must be in the format of ObjectName.`

`mxbeanInterface - the MXBean interface to be implemented by the proxy.`

Returns:

`a proxy for a platform MXBean interface of a given MXBean name that forwards its method calls through the given MBeanServerConnection, or null if not exist.`

Throws:

`IllegalArgumentException - if`

- `mxbeanName is not with a valid ObjectName format, or`
- `the named MXBean in the connection is not a MXBean provided by the platform, or`
- `the named MXBean is not registered in the MBeanServerConnection, or`
- `the named MXBean is not an instance of the given mxbeanInterface`

`IOException - if a communication problem occurred when accessing the MBeanServerConnection.`

getPlatformMXBean

```
public static <T extends PlatformManagedObject> T getPlatformMXBean(Class<T> mxbe
```

Returns the platform MXBean implementing the given `mxbeanInterface` which is specified to have one single instance in the Java virtual machine. This method may return `null` if the management interface is not implemented in the Java virtual machine (for example, a Java virtual machine

with no compilation system does not implement CompilationMXBean); otherwise, this method is equivalent to calling:

```
getPlatformMXBeans(mxbeanInterface).get(0);
```

Type Parameters:

T - an mxbeanInterface type parameter

Parameters:

mxbeanInterface - a management interface for a platform MXBean with one single instance in the Java virtual machine if implemented.

Returns:

the platform MXBean that implements mxbeanInterface, or null if not exist.

Throws:

IllegalArgumentException - if mxbeanInterface is not a platform management interface or not a singleton platform MXBean.

Since:

1.7

getPlatformMXBeans

```
public static <T extends PlatformManagedObject> List<T> getPlatformMXBeans(Class<
```

Returns the list of platform MXBeans implementing the given mxbeanInterface in the Java virtual machine. The returned list may contain zero, one, or more instances. The number of instances in the returned list is defined in the specification of the given management interface. The order is undefined and there is no guarantee that the list returned is in the same order as previous invocations.

Type Parameters:

T - an mxbeanInterface type parameter

Parameters:

mxbeanInterface - a management interface for a platform MXBean

Returns:

the list of platform MXBeans that implement mxbeanInterface.

Throws:

IllegalArgumentException - if mxbeanInterface is not a platform management interface.

Since:

1.7

getPlatformMXBean

```
public static <T extends PlatformManagedObject> T getPlatformMXBean(MBeanServerCo
                                                                     Class<T> mxbe
                                                              throws IOException
```

Returns the platform MXBean proxy for `mxbeanInterface` which is specified to have one single instance in a Java virtual machine and the proxy will forward the method calls through the given `MBeanServerConnection`. This method may return `null` if the management interface is not implemented in the Java virtual machine being monitored (for example, a Java virtual machine with no compilation system does not implement `CompilationMXBean`); otherwise, this method is equivalent to calling:

```
getPlatformMXBeans(connection, mxbeanInterface).get(0);
```

Type Parameters:

`T` - an `mxbeanInterface` type parameter

Parameters:

`connection` - the `MBeanServerConnection` to forward to.

`mxbeanInterface` - a management interface for a platform MXBean with one single instance in the Java virtual machine being monitored, if implemented.

Returns:

the platform MXBean proxy for forwarding the method calls of the `mxbeanInterface` through the given `MBeanServerConnection`, or `null` if not exist.

Throws:

`IllegalArgumentException` - if `mxbeanInterface` is not a platform management interface or not a singleton platform MXBean.

`IOException` - if a communication problem occurred when accessing the `MBeanServerConnection`.

Since:

1.7

See Also:

`newPlatformMXBeanProxy(javax.management.MBeanServerConnection, java.lang.String, java.lang.Class<T>)`

getPlatformMXBeans

```
public static <T extends PlatformManagedObject> List<T> getPlatformMXBeans(MBeanS
                                                                            Class<
                                                                    throws IOExce
```

Returns the list of the platform MXBean proxies for forwarding the method calls of the `mxbeanInterface` through the given `MBeanServerConnection`. The returned list may contain zero, one, or more instances. The number of instances in the returned list is defined in the specification of the given management interface. The order is undefined and there is no guarantee that the list returned is in the same order as previous invocations.

Type Parameters:

`T` - an `mxbeanInterface` type parameter

Parameters:

`connection` - the `MBeanServerConnection` to forward to.

`mxbeanInterface` - a management interface for a platform MXBean

Returns:

the list of platform MXBean proxies for forwarding the method calls of the `mxbeanInterface` through the given `MBeanServerConnection`.

Throws:

`IllegalArgumentException` - if `mxbeanInterface` is not a platform management interface.

`IOException` - if a communication problem occurred when accessing the `MBeanServerConnection`.

Since:

1.7

See Also:

`newPlatformMXBeanProxy(javax.management.MBeanServerConnection, java.lang.String, java.lang.Class<T>)`

getPlatformManagementInterfaces

```
public static Set<Class<? extends PlatformManagedObject>> getPlatformManagementIr
```

Returns the set of `Class` objects, subinterface of `PlatformManagedObject`, representing all management interfaces for monitoring and managing the Java platform.

Returns:

the set of Class objects, subinterface of PlatformManagedObject representing the management interfaces for monitoring and managing the Java platform.

Since:

1.7

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